Dense Dilated Network for Few Shot Action Recognition

Baohan Xu*
ZhongAn Technology
Shanghai, China
xubaohan@zhongan.io

Hao Ye*
Shanghai Advanced Research
Institute, CAS, China
yeh@sari.ac.cn

Yingbin Zheng
Shanghai Advanced Research
Institute, CAS, China
zhengyb@sari.ac.cn

Heng Wang
ZhongAn Technology
Shanghai, China
wangheng@zhongan.io

Tianyu Luwang†
ZhongAn Technology
Shanghai, China
luwangtianyu@zhongan.io

Yu-Gang Jiang
Fudan University
Shanghai, China
ygj@fudan.edu.cn

ABSTRACT

Recently, video action recognition has been widely studied. Training deep neural networks requires a large amount of well-labeled videos. On the other hand, videos in the same class share high-level semantic similarity. In this paper, we introduce a novel neural network architecture to simultaneously capture local and long-term spatial temporal information. The dilated dense network is proposed with the blocks being composed of densely-connected dilated convolutions layers. The proposed framework is capable of fusing each layer's outputs to learn high-level representations, and the representations are robust even with only few training snippets. The aggregations of dilated dense blocks are also explored. We conduct extensive experiments on UCF101 and demonstrate the effectiveness of our proposed method, especially with few training examples.

KEYWORDS

Action recognition; dilated convolution; few-shot learning; video analysis

ACM Reference Format:
Baohan Xu, Hao Ye, Yingbin Zheng, Heng Wang, Tianyu Luwang, and Yu-Gang Jiang. 2018. Dense Dilated Network for Few Shot Action Recognition. In *Proceedings of 2018 International Conference on Multimedia Retrieval, Yokohama, Japan, June 11–14, 2018 (ICMR '18)*, 9 pages.
https://doi.org/10.1145/3206025.3206028

1 INTRODUCTION

Video analysis is one of the most fundamental computer vision tasks, and has drawn a significant amount of attention with the popularity of video capture devices and the surge of Internet [10, 18, 19, 24, 28, 30]. Recently, deep neural networks achieved state-of-the-arts on many computer vision benchmarks, such as image classification and object detection [6, 8, 12, 25]. But the data volume for training the video analysis system is orders of magnitude larger than images. Annotating these large and varying kind of videos is labor intensive and time consuming due to their complexity and diversity. Therefore, the data scarcity problem is a common issue for newly emerged action categories. This impedes the neural networks from learning discriminative features. Figure 1 shows some example frames in video action datasets. The actions in the same category have variety of backgrounds, which requires sufficient amount of annotated video data for training a deep neural network. It would be better if we can find a way to learn new action categories with few examples.

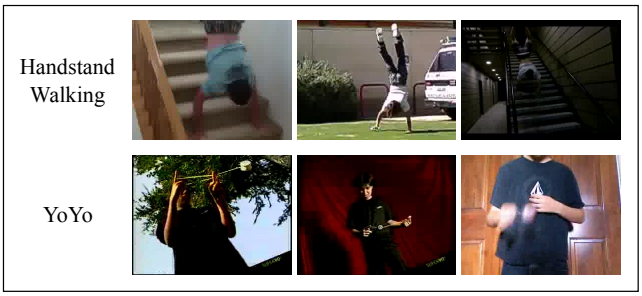

Figure 1: Some example frames of video action dataset UCF101 [26]. Although the same class of videos contain high-level semantic correlations, different backgrounds and lack of quality control makes few-shot learning more challenging.

Few-shot learning aims to fully make use of knowledges that have been obtained so far from pre-training rather than to start from scratch when new action categories need to be learned. A recent highly effective prototype-based learning paradigm on few-shot learning attempts to resolve the data scarcity problem by using a pre-trained deep neural network as a starting point [3]. In action recognition, many different kinds of actions share similar spatial or motion primitives, whereas a new classifier, referred to as prototype, may be reconstructed by taking these shared primitives and the additional examples into account. It is widely acknowledged that

*These authors contributed equally to this work.
†Corresponding author.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
ICMR '18, June 11–14, 2018, Yokohama, Japan
© 2018 Copyright held by the owner/author(s). Publication rights licensed to Association for Computing Machinery.
ACM ISBN 978-1-4503-5046-4/18/06…$15.00
https://doi.org/10.1145/3206025.3206028

deep neural networks provide a promising capability of keeping and reusing these primitives.

In this paper, we propose a dilated dense connection based deep network architecture that deals with the few-shot action recognition by framing the problem within the prototype learning settings. The densely connected blocks capture not only the spatial information within the layer-wise concatenation, but also the long-term context through the dilated temporal convolution. Rather than retraining the action recognizer by replacing the classifier in an incremental manner, we regard the existing networks trained on seen data as action primitive extractors. Our proposed networks take these primitives as inputs so that they can adapt to those never-before-seen actions easily.

In this dilated dense connection based framework, two basic ConvNets are pre-trained on frames and stacked optical flows respectively. The spatial and motion primitives extracted from these two pathways are sent to the consecutive densely connected architecture where each layer receives additional inputs from all preceding layers except the basic ConvNet and pass its features to the subsequent layers. Thereon, a compact flow of information is fully employed to aggregate the spatial and temporal features from the few training video snippets. Under this point of view, densely connected blocks and temporal dilated convolutions are capable of mining few action examples by unleashing the potential of densely connected blocks.

Our contributions are summarized as follows:

- We propose a novel deep neural network architecture combing dilated temporal convolution and densely layer connection to perform video action recognition. Our framework can generate new class prototypes where only a small number of new action videos are needed.
- The dilated dense blocks in our framework are able to capture spatial-temporal information from both snippet-level and long-term context.
- We achieve the state-of-the-art action recognition performance on UCF101 by only using part of the training videos and then reveal the effectiveness of dense connection blocks on few-shot action recognition.

The rest of our paper is arranged as follows. We begin with a brief review of literatures on action recognition and few shot learning in Section 2. In Section 3, we give the details of our framework structure, and implementation stages of the densely connection configuration. We show a variety of experimental results conducted on UCF-101 to demonstrate the effectiveness in Section 4. Section 5 concludes the paper.

2 RELATED WORKS

2.1 Video Action Recognition

Video action recognition has been widely studied during last few years. A detailed survey can be found in [2]. Early researchers focused on hand-crafted feature extractions. Various features have been explored, such as space-time interest points (STIP) [14], histogram of gradient (HoG) and histogram of optical flow (HOF) [15]. Knowledge from image domain is extended to capture the temporal information. Further more, Wang et al. proposed dense trajectory features to densely sample and track local patches from each field in a dense optical flow field [28]. However, these hand-crafted features still have some limitations for further improvements, especially when dealing with large scale of video data.

Recent years, convolutional neural networks gained more and more attention. It showed great success in image classification and detection. In 2012, AlexNet [12] won ImageNet Challenge [23] with convolutional neural network. With ConvNets becoming more popular in computer vision field, a number of attempts have been made to improve performance. Simonyan et al. [25] proposed a deeper network by adding more convolutional layers. More recently, He et al. [6] developed a residual function with reference to the layer inputs. The residual nets, which have a total of 152 layers, are still able to increase accuracy gained from greatly increased depth. Huang et al. proposed DenseNet [8], which focuses on feature reuse and learns a variation in the input of different layers. These state-of-the-art networks were designed for static image classification. Directly using these frameworks on video action recognition tasks will lose lots of discriminative information such as temporal contexts and motions.

To tackle these problems, Simonyan et al. [24] proposed a fusion network which combines both static frames and motion optical flows. Ye et al. [32] evaluated important implementation options on two-stream CNN. Competitive results were attained on popular video classification benchmarks, such as HMDB51 [13], UCF101 [26] and ActivityNet [7]. While treating videos as frames or optical flow images separately will leave the temporal relations not fully explored.

Many studies also paid attentions to how to use temporal information in order to achieve multi-modality recognition. Tran et al. [27] adopted 3D CNN (C3D) for joint learning spatial-temporal representations in video datasets. Later in [21], Qiu constructed 2D spatial convolutions and 1D temporal connections to stimulate 3D convolutions. The deep residual learning framework also designed for efficiently training a deeper neural network. Instead of using traditional convolutions, Lea et al. [29] explored temporal convolutions for action segmentation tasks. Encoder-decoder temporal convolutional network (TCN) and dilated TCN were used to capture long-range temporal patterns. However, these approaches still require large amount of data and resources to get a satisfactory result.

2.2 Few Shot Learning

In early studies, few shot video action recognition mainly focused on datasets like KTH and Weizmann [20, 31]. These datasets consist of actors performing scripted "atomic" actions, such as walking, jogging and running, which are easy to recognize using only a small percentage of the training data. Hidden markov model (HMM) combined with encoding methods demonstrated some effectiveness [22]. However, when facing large scale unconstrained data, the performance of these methods are not very encouraging.

Recently, some researchers transfered pretrained image domain models to video action recognition on unconstrained video datasets [3], such as UCF101 [26]. Since these videos are generated by users with little quality control, it makes few shot recognition more challenging. Li et al. mapped videos into low-dimensional feature space using spatial attention map [17]. These domain adaptive methods

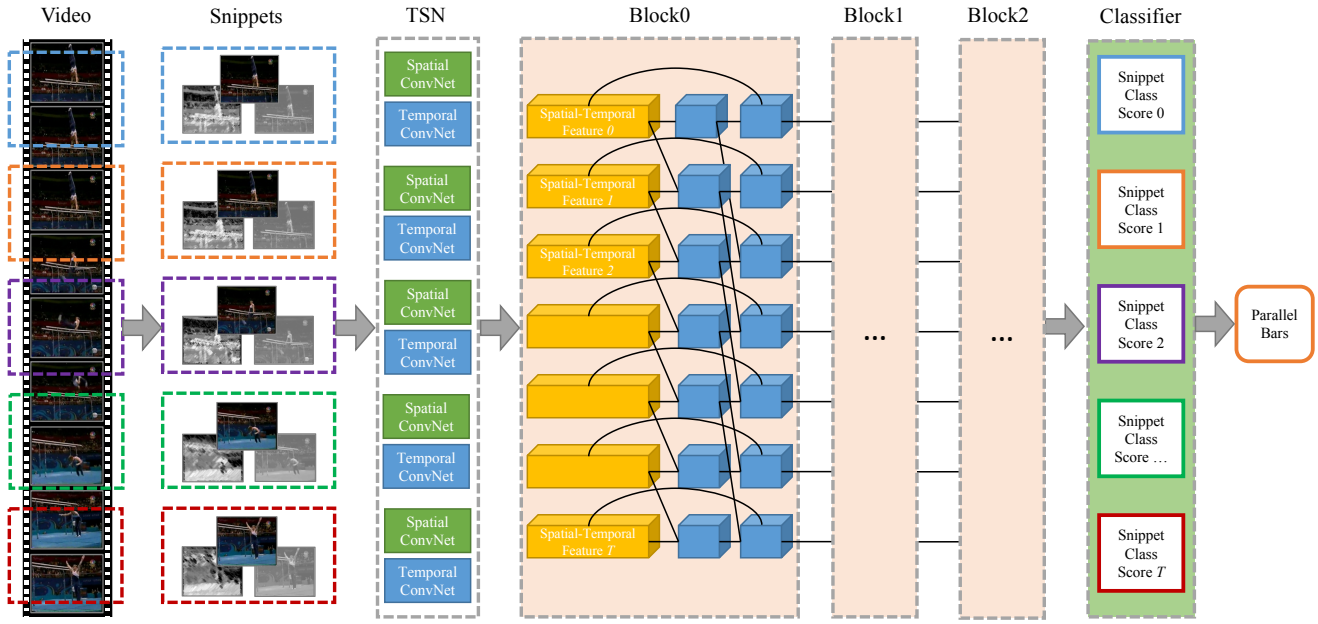

Figure 2: The framework of our proposed dense dilated network. Videos are divided into n snippets. Temporal segment network (TSN) is used to extract spatial-temporal features of every snippets. Dense dilated blocks are utilized to densely connect channel wise features. The dilated convolutions help explore temporal relations between different snippets. Prediction scores are made based on all the feature-maps in the network. The scores of all the snippets are leveraged to perform video-level prediction.

need auxiliary images or other heterogeneous information to transfer knowledge. Temporal and motion clues are neglected. Instead of using auxiliary information, few scholars focused on modifying the structure of the network to perform few-shot learning.

Inspired by previous researches, we propose a dense dilated temporal convolution network instead of constructing deep or complicated architectures. With only few training examples, actions in the same class could share some common high-level semantic attributions. We explore how to fully utilize local and global temporal information to learn inner-class semantic representations. First of all, the dilated temporal convolution provides wide receptive fields for each layer. Then concatenating different feature maps in each layer helps the model to learn long-range temporal representations, especially when the training data is limited. In addition, the densely connected architecture makes the whole network easier to train and more efficient.

3 METHOD

In this section, we introduce our proposed framework. Recent proposed DenseNet [8] is found useful in image classification tasks. Rather than traditional feed-forward architectures, DenseNet investigates a novel and simple connectivity pattern to allow maximum information flow between layers. The dense connections across channels help to achieve significant improvement of image tasks, especially on a small dataset.

Inspired by DenseNet, our proposed framework aims to fully utilize spatial-temporal information of the entire video using dense connections. As illustrated in Figure 2, for each snippet of videos, dense dilated blocks are designed to concatenate all feature maps, which provide both local and global spatial-temporal features to subsequent layers.

Formally, suppose we extracted n frames and optical flow images of each video. The temporal segment network (TSN) [29] is utilized to extract RGB features and flow features separately. The basic network of TSN is Google Inception network [9]. The TSN is trained on Kinetics dataset [11], which contains about 300,000 video action clips among 400 classes. Specifically, we use the 1024 dimensions of global pool features before the softmax layer.

The temporal dilated convolutions use different dilation parameters to get different views of temporal information, which has been shown to be effective in many tasks such as video action detection [16], semantic segmentation [1, 33], RGB-D [34], and DNA modeling [5]. Densely connection inside the blocks on channel level can give more variations to the input of subsequent blocks. Aggregation of each dense dilated block can leverage the global features of whole videos and the specific snippets, which is useful for recognition. We will explain each part in following subsections.

3.1 Dense Dilated Blocks

The advantage of DenseNet is not only to prevent gradient dispersion, but also to reduce the number of parameters. The layer of DenseNet is often very narrow, such as 12 filters per layer, which makes the whole network easy to train and converge. It's also worth noticing that dense connections have a regularizing effect [8]. It

can reduce overfitting, especially with smaller amount of training data. We propose dense dilated framework not only to improve video action recognition but also to enable few shot learning.

Figure 3 demonstrates the detailed information in each dilated convolutional block. We empirically set 3 layers as example, the performance of varying layer number are explored in Section 4.4. Every layer consists of three operations: a batch normalization [9], a dilated convolution, and a rectified linear uint (ReLU) [4] function.

Specifically, every block has a sequence of L layers. Every layer contains a batch normalization, a set of dilated convolutions with a rate parameter s, and a ReLU function. The dilation rate increases for consecutive layers in the block as $s_l = 2^l$. Compared to traditional convolutions, dilated convolution can capture long-range dependencies. This will give the discriminative snippets more weights to make contributions to the overall recognition. Convolutions are applied over two time steps, t and $t - s$. Thus the filters can be represented as $W = \{W^{(1)}, W^{(2)}\}$. Therefore, we can define the dilated convolution as

$$x_t^l = f(W^{(1)} x_{t-s}^{l-1} + W^{(2)} x_t^{l-1} + b), \tag{1}$$

where x_t^l refers to the dilated convolution results on time t in layer l and b is the bias vector.

To further improve our framework, we consider the dense connection within each block. Concatenating feature maps learned by different layers can provide more variations to the input of subsequent layers. Inner-class representations can be learned more easily, especially when the training data is limited. Layers in each temporal dilated convolution (TDC) block i have the same number of filters k^i, which could also be regarded as growth rate in DenseNet. The dense connection combines all layers in channel-wise manner.

More specifically, we define the output of every block B^i on time t as S_t^i,

$$S_t^i = [x_t^l, x_t^{(l-1)}, \ldots, x_t^0], \tag{2}$$

where S_t^i concatenate all the outputs into a single tensor.

Transition layers. We also add transition layers between blocks. It consists of a batch normalization layer followed by an 1×1 convolutional layer to down sample the feature maps.

Growth rate. We refer k as growth rate of different blocks. Each layer produce k^i feature-maps in the i-th block, then the filters of different blocks can write down as

$$k^i = k^{i-1} \times (l - 1) + k^0, \tag{3}$$

where k^0 refer to the number of channels in the input layer, l is the number of layers in each block. On limited training data, we show that a relatively small growth rate and small number of layers and blocks are sufficient to get a high accuracy.

3.2 Dense Dilated Aggregation

We explore different connection structures to build dense dilated network, we set 3 blocks as the basic structure of all the aggregations. Figure 4 illustrates 3 aggregation strategies, including adding all the TDC blocks, concatenating blocks and extracting features of different blocks. Utilize the information of all the blocks may provide more knowledge of the whole video, while the output of blocks from a simpler network could give different scales of information. We'll explain each way in detail as follows.

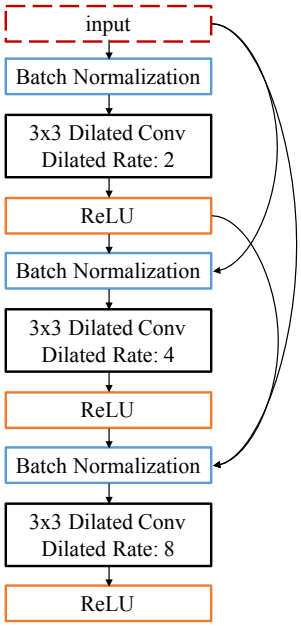

Figure 3: The framework of temporal dense convolution with 3 layers. Every layer consists of a batch normalization, a dilated convolution, and an activation function. Different layers are concatenated in a dense way.

Adding layers Naturally, we first tried adding the output of all layers together. Due to different shapes of filters in different blocks, we apply 1×1 convolutions to align all the output tensors. The output layer can be defined as:

$$Z_t = ReLU(V \sum_{i=0}^{B-1} S_t^i + e), \tag{4}$$

where V is the weight matrix and e is the bias. We refer this architecture as dense dilated adding (DDA) aggregation as (a) in Figure 4.

Concatenating layers Further more, in order to make receptive fields wider, we explore concatenating the output of different layers. The final layer has wider range of channels than other connection methods.

$$Z_t = ReLU([S_t^{B-1}, S_t^{B-2}, \ldots, S_t^0] + e). \tag{5}$$

This way we call it dense dilated concatenation (DDC) connection as (b) in Figure 4.

Output features of different blocks More straightforwardly, we extracted the output of different blocks to perform recognition of video actions. Considering the output as snippet-level representations helps us get more insights of the proposed architecture. This will also provide useful information when tuning the hyper-parameters. As illustrate as (c) in Figure 4, we refer the output features of each block as DD1, DD2, and DD3 respectively.

Compared to extracting features, the adding and concatenating ways maybe redundant, which produce unnecessary information not only make the network more complicated, but also make it harder to converge on small datasets.

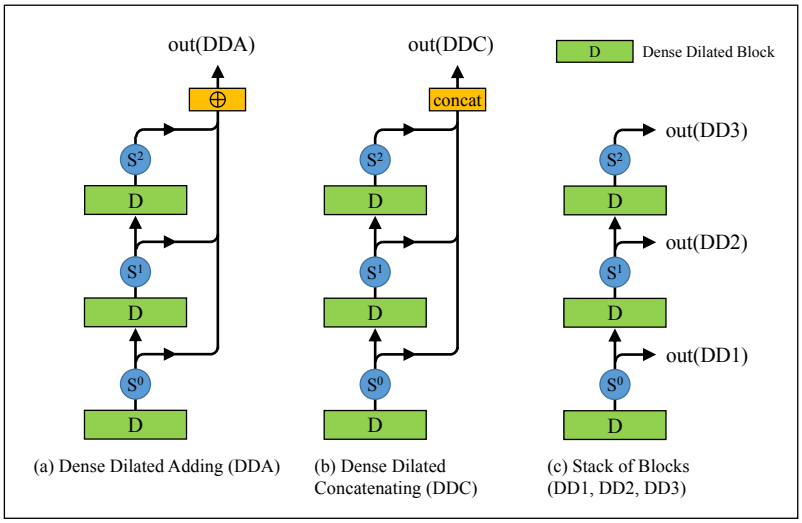

Figure 4: Different layer aggregation strategies. (a) is summed using a set of skip connections. (b) is concatenated over the output of different output blocks. (c) are more straightforward to extract every block as snippet-level representation.

3.3 Classification

We extract the last layer before softmax as output of all dense dilated networks. The output consists of n snippets $\times$ d-dimension features. A support vector machine (SVM) is used as the predictor $\mathcal{P}(\cdot)$ to predict the class label for every snippet. Majority vote is applied to decide the video-level prediction.

4 EVALUATIONS

In this section, we first introduce the experimental settings, and then validate the effectiveness of our framework using both whole data and few shot settings. Varying the hyper parameters are also fully explored in our experiments.

4.1 Dataset and Settings

We utilize popular video action recognition dataset UCF101 [26]. It consists of 101 action classes, 13,320 clips and 27 hours of unconstrained videos collected from YouTube. The large variations in camera motion, different background and illumination conditions makes it a challenge video action dataset. Figure 5 shows some example frames of different categories in UCF101. In order to enable few shot learning, we randomly sample 10%, 20%, 33%, and 50% videos from the training set. The experiments are separately done on the three public train/test splits for comparision, and the result is averaged across three test splits.

We extract global pool features from temporal segment network (TSN) with pretrained model on Kinetics. Each video contains 25 snippets as the default setting in TSN. RGB and Optical flow features are concatenated as snippet-level features for every video. The concatenating spatio-temporal features are regarded as the input of our frameworks. The growth rate K of all the block is set as 12 to leverage the accuracy and the computation cost.

Figure 5: Example frames of UCF101.

As illustrated in Figure 4 and Section 3.2, we compare some state-of-the-art methods as baselines with 3 different layer aggregation frameworks that we proposed.

4.2 Video Action Recognition

To evaluate our algorithm, we compare with different video action recognition methods. First four baseline methods make use of spatial-temporal features, which is similar to our method. **(1) C3D [27].** This method proposes an effective approach for spatial-temporal feature learning using deep 3-dimensional convolutional networks. **(2) P3D [21].** Spatial and temporal information in videos are combined to build a bottleneck block, which can simulate 3D

Table 1: Comparison with state-of-the-art methods on UCF101 using all data. The result of dilated TCN is computed by ourself, while other baseline results are reported in the papers.

Method	Accuracy(%)
C3D[27]	85.2
P3D[21]	88.60
Temporal Segment Network[29]	93.50
Dilated TCN[16]	92.31
EnergyNet[17]	88.00
DDA	94.74
DDC	95.30
DD1	95.17
DD2	**96.15**
DD3	95.48

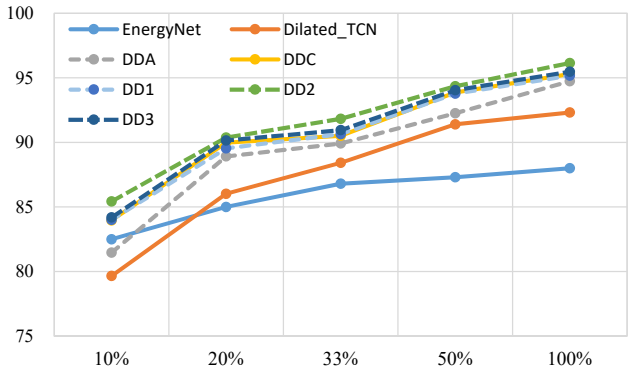

Figure 6: Few shot video action recognition results.

Table 2: Few shot results with only 20% training videos.

Method	Accuracy(%)
Dilated TCN [16]	86.02
EnergyNet [17]	85.00
DDA	88.92
DDC	89.98
DD1	89.54
DD2	**90.38**
DD3	90.15

Table 3: The accuracy of DDA and DDC with 2 dense dilated blocks using 20% training data and 100% training data.

Method	20%	100%
DDA	89.56	95.47
DDC	**90.45**	**95.76**

convolutions in an effective way. **(3) Temporal segment network [29].** This method combines a sparse temporal sampling strategy and video-level supervision to enable efficient and effective learning using the whole action video. **(4) Dilated TCN [16].** Dilated TCN is most related to our method. Residual connections and dilated convolutions are used to perform video segmentation and detection. **(5) EnergyNet [17].** While EnergyNet use web images to learn attention maps, which is also helpful in few shot learning. We'll discuss more detail in Section 4.3.

Table 1 shows the results for video action recognition using whole training set. It clearly shows that all our proposed frameworks outperform state-of-the-art methods on UCF101 when using all data. These results demonstrate that adding dense connection can not only retrieve features from action snippets but also capture high-level semantics of the whole video. The improvement of DDC over DDA shows that concatenating channels expands the reception fields, which helps action recognition. While among the three feature extraction frameworks, DD2 surprisingly achieves best performance. DD1 performs even worse than DDC. This indicates that the representation of two dense dilated blocks are enough for our task. The pooling operation between DD2 and DD3 may lose some significant spatial-temporal features.

4.3 Few Shot Recognition

We mainly compare our few shot learning methods with [17], which reported few shot results using auxiliary image datasets. They proposed an EnergyNet transferring knowledge from web images to videos. The transfer learning method can adapt to the target domain with limited amount of training data. However, EnergyNet only considers the attention information of static images. This framework is difficult to be expanded to use multi-modal information. While our framework enables the network to learn high-level semantics of each class without auxiliary images or other information. Image and optical flow are both utilized in our framework as multi-modal features.

The comparison results are reported in Table 2. It suggests that our dense dilated methods outperform our baseline methods using 20% training data. Figure 6 illustrates the accuracy when different amounts of training videos are used. DD2 still outperforms all other methods. Our proposed frameworks can also achieve better performance using fewer videos compared with the baseline methods. Using only 20% training videos, they all outperform EnergyNet which uses 100% training data. While with 50% videos, DD2 outperforms dilated TCN with whole training data. One interesting finding is that dense dilated methods get more improvements with less training data. Using 10% training videos, DD2 outperforms dilated TCN by 4.54%, compared with 2.46% when using 50% training data. This finding demonstrates our proposed method can preserve more intra-class relationship than baseline methods giving little training data.

To further evaluate the complimentary of different blocks, we also conduct experiments on DDA and DDC using 2 dense dilated blocks. Table 3 reported the accuracy for each architecture. The result of DDC outperforms other dense dilated structures, which proves the complementary of first 2 blocks. It also shows that adding the third block may not be a good choice.

Figure 7: Quantitive examples of successful and failure examples. 20% training data are used by DD2 model.

Figure 7 demonstrates some quantitive examples for successful and failure cases. The top four examples are the successfully recognized examples. Even with few training data, our methods can successfully classify most of the classes with different backgrounds and qualities. The green frames refer to the snippets that our frameworks are more confident, which also illustrates that the proposed method can identify important movements for recognition. The last two rows show some failure examples. The similar backgrounds and the unclear movements make the system misclassified these videos into Haircut and ThrowDiscus.

4.4 Hyper Parameter Tuning

The hyper-parameters in our frameworks include the number of layers and the growth rate. The changing of these hyper-parameters may affect the performance. In this subsection, we explore the influence of different choices of these settings. 20% training data are used in all the experiments.

Varying different layers. We compare the performance of our frameworks with varying number of layers L in the dense dilated block. In Figure 8, lines show accuracy for L from 2 to 4 . All the methods perform best with $L = 3$. Compared to $L = 2$, more layers in each block can boost the performance. However, when L gets

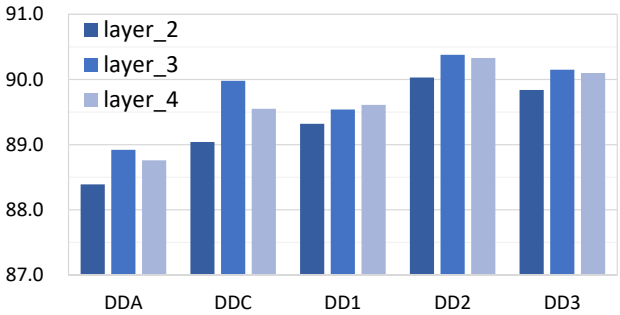

Figure 8: Performance of varying layer count L.

larger, the performances for both algorithms are decreased. This is because complicated neural networks lead to under fitting when the scale of training data is small.

Varying growth rate. Figure 9 shows how different growth rates affect the accuracy. Due to limited data, our dense dilated architecture can converge with relatively small growth rate. When $K = 12$ and $K = 16$, we can achieve a relatively high accuracy without drastically increasing the number of parameters for all the frameworks.

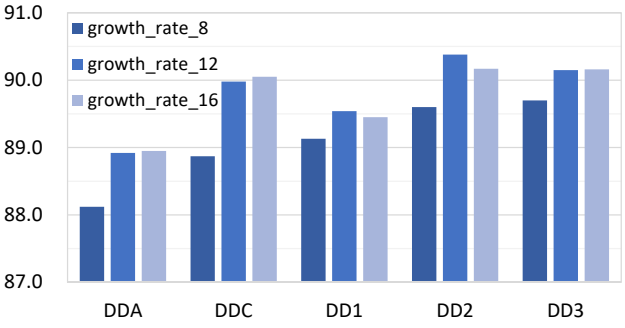

Figure 9: Performance of varying growth rate K.

The influence of Kinetics. To further validate whether the performance of our proposed methods is affected by the original training dataset Kinetics [11], we add another simple experiment to recognize the classes in UCF101 but not in Kinetics. According to manual visual inspection, about 45 classes in UCF101 are not the same as the classes in Kinetics. We use DD2 to train 20% of training videos in the 45 classes and do the evaluation. The accuracy is 86.23% of 45 classes. This shows our model can recognize video categories other than those in the pretrained models.

5 CONCLUSIONS

We introduce a novel dense dilated framework to perform video action recognition and few shot learning. The receptive fields include both local and global features via dense connections. The experiments show the effectiveness of our proposed framework compared to baseline metods. Our proposed framework also achieves state-of-the-art performance when facing limited data. We further explore the different strategies for layer aggregation among dense dilated blocks. The concatenating way is better than adding up all the blocks. The features of the second layer of blocks achieve best results under all conditions.

For further improvements, we will explore learning from scratch to perform video action recognition on more variety of videos. Video segmentation and action detection are also important topics that we leave for further investigation.

ACKNOWLEDGMENTS

This work was supported in part by the National Natural Science Foundation of China under Grants 61602459, 61622204, and 61572134, and in part by the Science and Technology Commission of Shanghai Municipality under Grant 16QA1400500.

REFERENCES

[1] Liang-Chieh Chen, George Papandreou, Iasonas Kokkinos, Kevin Murphy, and Alan L Yuille. 2017. Deeplab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected crfs. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2017).

[2] Guangchun Cheng, Yiwen Wan, Abdullah N Saudagar, Kamesh Namuduri, and Bill P Buckles. 2015. Advances in human action recognition: A survey. *arXiv:1501.05964* (2015).

[3] Chuang Gan, Chen Sun, Lixin Duan, and Boqing Gong. 2016. Webly-supervised video recognition by mutually voting for relevant web images and web video frames. In *European Conference on Computer Vision (ECCV)*. 849–866.

[4] Xavier Glorot, Antoine Bordes, and Yoshua Bengio. 2011. Deep sparse rectifier neural networks. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*. 315–323.

[5] Ankit Gupta and Alexander M Rush. 2017. Dilated Convolutions for Modeling Long-Distance Genomic Dependencies. *arXiv:1710.01278* (2017).

[6] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 770–778.

[7] Fabian Caba Heilbron, Victor Escorcia, Bernard Ghanem, and Juan Carlos Niebles. 2015. Activitynet: A large-scale video benchmark for human activity understanding. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 961–970.

[8] Gao Huang, Zhuang Liu, Kilian Q Weinberger, and Laurens van der Maaten. 2016. Densely connected convolutional networks. *arXiv:1608.06993* (2016).

[9] Sergey Ioffe and Christian Szegedy. 2015. Batch normalization: Accelerating deep network training by reducing internal covariate shift. In *International Conference on Machine Learning (ICML)*. 448–456.

[10] Yu-Gang Jiang, Yanran Wang, Rui Feng, Xiangyang Xue, Yingbin Zheng, and Hanfang Yang. 2013. Understanding and Predicting Interestingness of Videos. In *AAAI Conference on Artificial Intelligence (AAAI)*.

[11] Will Kay, Joao Carreira, Karen Simonyan, Brian Zhang, Chloe Hillier, Sudheendra Vijayanarasimhan, Fabio Viola, Tim Green, Trevor Back, Paul Natsev, Mustafa Suleyman, and Andrew Zisserman. 2017. The Kinetics Human Action Video Dataset. *arXiv:1705.06950* (2017).

[12] Alex Krizhevsky, Ilya Sutskever, and Geoffrey E Hinton. 2012. ImageNet classification with deep convolutional neural networks. In *Neural Information Processing Systems (NIPS)*.

[13] Hildegard Kuehne, Hueihan Jhuang, Estíbaliz Garrote, Tomaso Poggio, and Thomas Serre. 2011. HMDB: a large video database for human motion recognition. In *International Conference on Computer Vision (ICCV)*. 2556–2563.

[14] Ivan Laptev. 2005. On space-time interest points. *International Journal of Computer Vision* 64, 2-3 (2005), 107–123.

[15] Ivan Laptev, Marcin Marszalek, Cordelia Schmid, and Benjamin Rozenfeld. 2008. Learning realistic human actions from movies. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

[16] Colin Lea, Michael Flynn, Rene Vidal, Austin Reiter, and Greg Hager. 2017. Temporal Convolutional Networks for Action Segmentation and Detection. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

[17] Junnan Li, Yongkang Wong, Qi Zhao, and Mohan S Kankanhalli. 2017. Attention Transfer from Web Images for Video Recognition. In *ACM International Conference on Multimedia (MM)*.

[18] Yao Lu, Aakanksha Chowdhery, and Srikanth Kandula. 2016. Optasia: A relational platform for efficient large-scale video analytics. In *ACM Symposium on Cloud Computing (SoCC)*.

[19] Joe Yue-Hei Ng, Matthew Hausknecht, Sudheendra Vijayanarasimhan, Oriol Vinyals, Rajat Monga, and George Toderici. 2015. Beyond Short Snippets: Deep Networks for Video Classification. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

[20] Carlos Orrite, Mario Rodríguez, and Miguel Montañés. 2011. One-sequence learning of human actions. In *International Workshop on Human Behavior Understanding*. Springer, 40–51.

[21] Zhaofan Qiu, Ting Yao, and Tao Mei. 2017. Learning spatio-temporal representation with pseudo-3d residual networks. In *International Conference on Computer Vision (ICCV)*. 5534–5542.

[22] Mario Rodriguez, Carlos Orrite, Carlos Medrano, and Dimitrios Makris. 2017. Fast Simplex-HMM for one-shot learning activity recognition. In *CVPR Workshops*. 41–48.

[23] Olga Russakovsky, Jia Deng, Hao Su, Jonathan Krause, Sanjeev Satheesh, Sean Ma, Zhiheng Huang, Andrej Karpathy, Aditya Khosla, Michael Bernstein, Alexander C. Berg, and Li Fei-Fei. 2015. ImageNet Large Scale Visual Recognition Challenge. *International Journal of Computer Vision* 115, 3 (2015), 211–252.

[24] Karen Simonyan and Andrew Zisserman. 2014. Two-stream convolutional networks for action recognition in videos. In *Neural Information Processing Systems (NIPS)*. 568–576.

[25] Karen Simonyan and Andrew Zisserman. 2015. Very Deep Convolutional Networks for Large-Scale Image Recognition. In *International Conference on Learning Representations (ICLR)*.

[26] Khurram Soomro, Amir Roshan Zamir, and Mubarak Shah. 2012. UCF101: A dataset of 101 human actions classes from videos in the wild. *arXiv:1212.0402* (2012).

[27] Du Tran, Lubomir Bourdev, Rob Fergus, Lorenzo Torresani, and Manohar Paluri. 2015. Learning spatiotemporal features with 3d convolutional networks. In *International Conference on Computer Vision (ICCV)*. 4489–4497.

[28] Limin Wang, Yu Qiao, and Xiaoou Tang. 2015. Action recognition with trajectory-pooled deep-convolutional descriptors. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*. 4305–4314.

[29] Limin Wang, Yuanjun Xiong, Zhe Wang, Yu Qiao, Dahua Lin, Xiaoou Tang, and Luc Van Gool. 2016. Temporal segment networks: Towards good practices for deep action recognition. In *European Conference on Computer Vision (ECCV)*. 20–36.

[30] Baohan Xu, Yanwei Fu, Yu-Gang Jiang, Boyang Li, and Leonid Sigal. 2016. Video emotion recognition with transferred deep feature encodings. In *ACM International Conference on Multimedia Retrieval (ICMR)*. 15–22.

[31] Yang Yang, Imran Saleemi, and Mubarak Shah. 2013. Discovering motion primitives for unsupervised grouping and one-shot learning of human actions, gestures, and expressions. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 35, 7 (2013), 1635–1648.

[32] Hao Ye, Zuxuan Wu, Rui-Wei Zhao, Xi Wang, Yu-Gang Jiang, and Xiangyang Xue. 2015. Evaluating two-stream CNN for video classification. In *ACM International Conference on Multimedia Retrieval (ICMR)*. 435–442.

[33] Fisher Yu, Vladlen Koltun, and Thomas Funkhouser. 2017. Dilated residual networks. In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.

[34] Yingbin Zheng, Hao Ye, Li Wang, and Jian Pu. 2018. Learning Multiviewpoint Context-Aware Representation for RGB-D Scene Classification. *IEEE Signal Processing Letters* 25, 1 (2018), 30–34.